Bellabeat Smart Device Usage Analysis

Understanding User Behaviour and Developing Marketing Recommendations

1. Introduction

Business Task

Bellabeat is a high-tech manufacturer of health-focused products designed to help women better understand and improve their health. As a junior data analyst on the marketing analytics team, the objective of this project was to analyze smart device usage data to understand how consumers interact with fitness tracking devices.

The goal was to identify user behaviour patterns and provide insights that can help Bellabeat improve marketing strategies, increase user engagement, and encourage healthier lifestyles.

The analysis focused on answering:

- How frequently do users interact with their devices?
- When are users most and least active?
- How are activity levels related to calories, sleep, and sedentary behaviour?
- What types of users should Bellabeat target?

2. Data Source

The analysis used the Fitbit Fitness Tracker Dataset, which contains health and activity data collected from multiple users through wearable fitness devices.

Source: Kaggle
Dataset: Fitbit Fitness Tracker Data
Link: FitBit Fitness Tracker Data

The original datasets included:

Daily Activity Dataset

Contained daily measurements of: User ID, Total steps, Distance travelled, Calories burned, Sedentary minutes, Lightly active minutes, Fairly active minutes, Very active minutes

Sleep Dataset

Contained sleep-related information:User ID, Total minutes asleep, Total time in bed

Hourly Activity Datasets

Hourly datasets were used to understand activity behaviour throughout the day.

These included: Hourly steps, Hourly intensity levels, Hourly activity records

3. Data Preparation and Cleaning Using SQL

The raw datasets were imported into SQL Server Management Studio (SSMS) for cleaning, transformation, and analysis.

3.1 Data Quality Checks

The following checks were performed:

Checking Dataset Structure

Reviewed:

- Number of rows
- Number of columns
- Column names
- Data types

This helped understand the structure of each dataset before analysis.

Checking Missing Values

Missing values were checked across datasets to ensure that important variables required for analysis were complete.

Checking Duplicate Records

Duplicate records were identified and removed where necessary to prevent inaccurate calculations.

Checking Data Consistency

The following were reviewed: Date formats, User IDs, Numerical fields, Activity measurements

Data types were corrected where necessary to ensure successful joins and calculations.

4. Data Integration and Transformation

After cleaning, multiple datasets were combined to create analysis-ready tables.

4.1 Activity and Sleep Integration

The daily activity dataset was joined with the sleep dataset using: **User ID**

This created: ActivitySleep_Clean

This table was used for:

- KPI calculations
- Activity vs calories analysis
- Activity vs sleep analysis
- Sedentary behaviour analysis
- User segmentation

4.2 Hourly Activity Integration

The hourly datasets were combined:

- Hourly Steps
- Hourly Intensities
- Hourly Activity Data

This created: hourlyActivity_clean_V2

This table was used for:

- Hourly activity trends
- Identifying peak activity periods
- Identifying low activity periods

5. Feature Engineering

Additional categories were created using SQL and Tableau calculated fields.

Activity Classification

Users were grouped based on daily steps:

Category	Criteria
Sedentary	Less than 5,000 steps
Moderately Active	5,000–9,999 steps
Highly Active	10,000+ steps

Sedentary Classification

Sedentary behaviour was grouped into:

- Low Sedentary
- Moderate Sedentary
- High Sedentary

This helped understand how inactivity affects overall movement.

6. Tableau Analysis and Visualization

The cleaned datasets were imported into Tableau to create interactive dashboards and a story.

The analysis was divided into two dashboards:

Dashboard 1: User Activity Overview

KPI Overview

The overall user averages were:

Metric	Average

Daily Steps	7,638
Calories Burned	2,300
Sleep Duration	417 minutes
Sedentary Time	991 minutes

Insight

Users were moderately active but spent a considerable amount of time sedentary.

Weekly Activity Pattern

Findings

- Saturday recorded the highest activity.
- Sunday recorded the lowest activity.

Insight

Users were more engaged with physical activity during certain days of the week, particularly weekends.

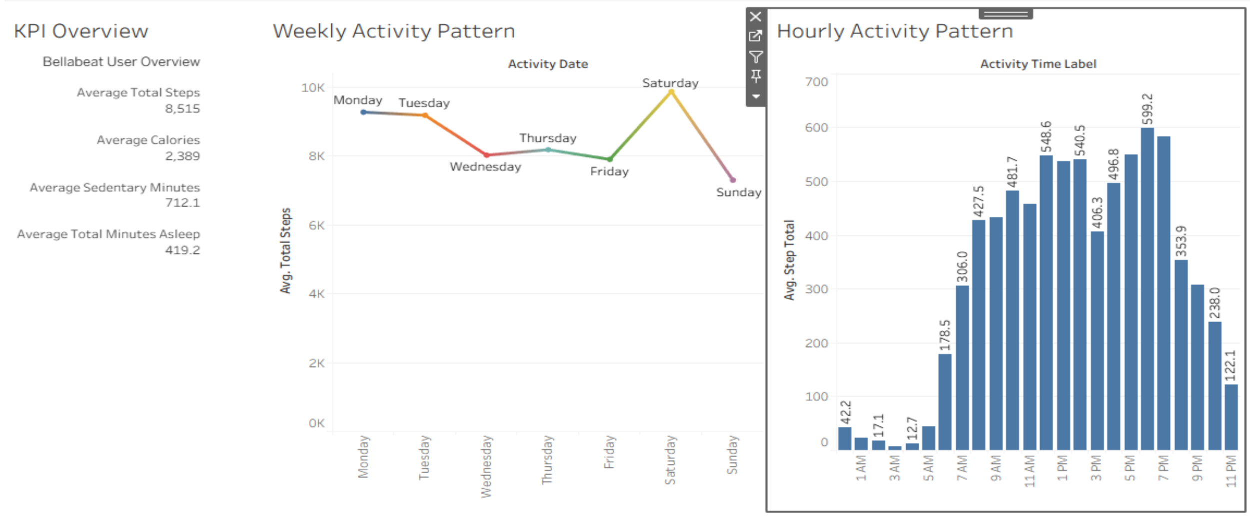
Hourly Activity Pattern

Findings

- Peak activity occurred around **6 PM**.
- Strong activity was observed between **5 PM and 7 PM**.
- Lowest activity occurred around **3 AM**.

Insight

Users have predictable daily activity periods that can guide engagement strategies.



Dashboard 2: User Behaviour and Health Insights

Activity vs Calories

Finding

Users with higher activity levels burned more calories.

Insight

Physical activity is directly associated with increased calorie expenditure.

Activity vs Sleep

Finding

Users showed different sleep patterns depending on activity levels.

Insight

Combining activity and sleep insights allows Bellabeat to position itself as a complete wellness platform.

Sedentary Behaviour vs Activity

Finding

Users with higher sedentary time generally recorded fewer daily steps.

Insight

Reducing sedentary behaviour could encourage increased physical activity.

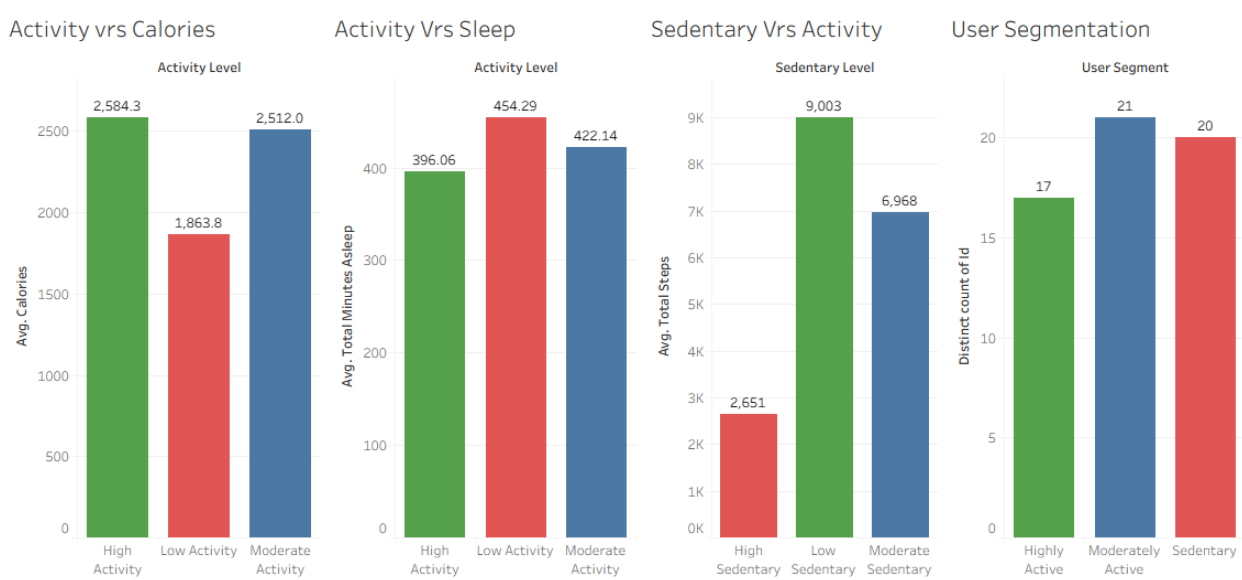
User Segmentation

Users were categorized as:

Category	Number of Users
Sedentary	20
Moderately Active	21
Highly Active	17

Insight

Most users were sedentary or moderately active, creating an opportunity for Bellabeat to improve engagement among these groups.



7. Recommendations

Recommendation 1: Re-engage Users During Low Activity Periods

Evidence:

- Sunday had the lowest activity levels.
- Activity was lowest around 3 AM.

Action:
Bellabeat should use personalized reminders and wellness prompts during low-engagement periods to encourage consistent activity.

Recommendation 2: Leverage Peak Activity Periods

Evidence:

- Saturday had the highest activity.
- Activity peaked around 6 PM, especially between 5 PM and 7 PM.

Action:
Bellabeat should schedule motivational notifications, challenges, and achievement messages during peak activity periods.

Recommendation 3: Encourage Consistent Weekly Activity

Evidence:

- Activity levels varied throughout the week, with lower engagement on Sunday.

Action:
Bellabeat should introduce weekly goals, progress tracking, and challenges to maintain user engagement.

Recommendation 4: Use Activity Achievements to Motivate Users

Evidence:

- Highly active users burned more calories and showed lower sedentary behaviour.

Action:
Bellabeat should highlight achievements such as calories burned, milestones reached, and progress improvements.

Recommendation 5: Target Sedentary and Moderately Active Users

Evidence:

- Sedentary users: 20
- Moderately Active users: 21
- Highly Active users: 17

Action:
Bellabeat should focus on personalized goals, reminders, and challenges to gradually increase activity among less active users.

8. Conclusion

The analysis showed that Fitbit users demonstrate different activity and wellness behaviours. While some users maintain high activity levels, many users fall within sedentary or moderately active groups.

Bellabeat can improve engagement by using personalized recommendations, optimizing notification timing, and promoting its products as complete wellness solutions that combine activity, calorie, and sleep tracking.

The findings provide actionable insights that can support Bellabeat's marketing strategy and increase long-term user engagement.

Tools Used

- SQL Server Management Studio (Data cleaning, joining, and analysis)
- Tableau (Dashboards and storytelling)
- Excel/CSV files (Data handling)

Appendix A: SQL Queries

1. Data Cleaning: Checking Duplicates

```
SELECT
    Id,
    ActivityDate,
    COUNT(*) AS duplicate_count
FROM ActivitySleep
GROUP BY
    Id,
    ActivityDate
HAVING COUNT(*) > 1;
```

2. Creating Activity and Sleep Summary Table

```
SELECT
    a.Id,
    a.ActivityDate,
    a.TotalSteps,
    a.Calories,
    a.SedentaryMinutes,
    s.TotalMinutesAsleep,
    s.TotalTimeInBed
FROM DailyActivity a
LEFT JOIN SleepData s
ON a.Id = s.Id
AND a.ActivityDate = s.SleepDay;
```

3. User Activity Segmentation

```
SELECT
    CASE
        WHEN TotalSteps < 5000 THEN 'Sedentary'
        WHEN TotalSteps BETWEEN 5000 AND 9999 THEN 'Moderately Active'
        ELSE 'Highly Active'
    END AS User_Category,

    COUNT(DISTINCT Id) AS Number_of_users

FROM ActivitySleep_Clean
```

```
GROUP BY
    CASE
        WHEN TotalSteps < 5000 THEN 'Sedentary'
        WHEN TotalSteps BETWEEN 5000 AND 9999 THEN 'Moderately Active'
        ELSE 'Highly Active'
    END;
```

4. Sedentary Behaviour Analysis

```
SELECT
    CASE
        WHEN SedentaryMinutes < 800 THEN 'Low Sedentary'
        WHEN SedentaryMinutes BETWEEN 800 AND 1200 THEN 'Moderate
Sedentary'
        ELSE 'High Sedentary'
    END AS Sedentary_Category,

    AVG(TotalSteps) AS Average_Steps

FROM ActivitySleep_Clean

GROUP BY
    CASE
        WHEN SedentaryMinutes < 800 THEN 'Low Sedentary'
        WHEN SedentaryMinutes BETWEEN 800 AND 1200 THEN 'Moderate
Sedentary'
        ELSE 'High Sedentary'
    END;
```

5. Hourly Activity Pattern

```
SELECT
    ActivityHour,
    AVG(TotalSteps) AS Average_Steps,
    AVG(TotalIntensity) AS Average_Intensity
FROM hourlyActivity_clean_V2
GROUP BY ActivityHour
ORDER BY ActivityHour;
```

6. Weekly Activity Pattern

```
SELECT
   DATENAME(WEEKDAY, ActivityDate) AS Day_of_Week,
```

```
    AVG(TotalSteps) AS Average_Steps,
    AVG(Calories) AS Average_Calories
FROM ActivitySleep_Clean
GROUP BY DATENAME(WEEKDAY, ActivityDate);
```